RF vs SVM for Real-Time Intrusion Detection: A Comparative Evaluation with Ablation and Drift Analysis

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***Abstract*—Machine learning-based Intrusion Detection Systems (IDS) must process highly imbalanced network traffic while meeting strict latency constraints for real-time response. Existing studies compare models on accuracy metrics without jointly evaluating inference cost, resilience to dimensionality reduction, and robustness to concept drift. This study presents a comparative evaluation between Random Forest (RF) and Support Vector Machine (SVM) under a latency constraint of 50 ms, including an ablation study and robustness analysis under synthetic drift. Experiments are conducted on CIC-IDS2017 (800,000 samples, 78 features) with validation over $k = 5$ independent runs, reporting mean $\pm$ standard deviation. RF achieves PR-AUC of 0.9997 ± 0.0000 versus 0.9196 ± 0.0016 for LinearSVM in static conditions. Under ablation to 10 features, RF maintains PR-AUC of 0.9962 while SVM-RBF drops to 0.6434. However, under concept drift, RF collapses rapidly (F1 -98.3% at 5% drift) while LinearSVM degrades gracefully (F1 -7.9%), revealing a precision-robustness tradeoff where model choice depends on operational stability requirements.**

***Index Terms*—Intrusion Detection, Random Forest, SVM, Real-Time, Ablation Study, Concept Drift, CIC-IDS2017**

I. Introduction

Modern computer networks face increasingly sophisticated cyberattacks, requiring Intrusion Detection Systems (IDS) capable of identifying malicious behavior in real-time. Machine learning approaches have demonstrated effectiveness for this task, with Random Forest and SVM identified as the most effective approaches on classical benchmarks [1]. These systems must satisfy multiple simultaneous constraints: high detection accuracy, particularly for minority classes representing attacks, and sufficiently low inference latency to enable rapid response before damage occurs.

Network traffic exhibits significant class imbalance between normal traffic (majority) and attacks (minority). On the CIC-IDS2017 dataset [2], this ratio reaches 80.75% benign traffic versus 19.25% attacks. This imbalance renders classical metrics like accuracy inappropriate, as a classifier predicting all samples as benign would achieve 80% accuracy while detecting zero attacks. For this reason, PR-AUC (Area Under Precision-Recall Curve) is preferred for evaluation on imbalanced datasets [3]. The latency constraint, typically set below 50 ms for real-time applications based on network response requirements [4], limits the choice of deployable algorithms and excludes computationally expensive models such as deep neural networks.

Despite recent advances, three limitations persist in the literature. First, most comparative studies focus on accuracy without systematically measuring inference latency, making it unclear whether reported models are deployable in production. Second, model resilience to drastic dimensionality reduction (feature ablation) remains insufficiently studied, yet this is critical for resource-constrained environments such as IoT devices [6]. Third, few works evaluate robustness to concept drift, a common phenomenon when attack signatures evolve over time [7].

To address these gaps, we formulate the following research question:

- **RQ1**: Under strict latency constraints (<50 ms), which model family (ensemble vs linear) maximizes detection capability (PR-AUC) on imbalanced data, and how does this tradeoff evolve under feature ablation and concept drift?

The main contributions of this work are:

1) A comparative evaluation of RF vs SVM with statistical validation (mean $\pm$ std over $k = 5$ runs), jointly measuring PR-AUC, Recall, F1-Score, and inference latency under the same experimental conditions.
2) An ablation study quantifying the impact of feature reduction from 78 to 10 features on each model family, demonstrating RF's superior resilience.
3) A robustness analysis under synthetic concept drift (0% to 40%), revealing an unanticipated tradeoff between precision and stability where RF excels statically but SVM degrades more gracefully.
4) A reproducible experimental protocol with code available on GitHub[1].

The remainder of this paper is organized as follows. Section II reviews related work and positions our contribution. Section III describes the experimental methodology including dataset, preprocessing, and evaluation protocol. Section IV reports the obtained results. Section V discusses implications, the precision-robustness tradeoff, and limitations. Section VI concludes and outlines future work.

[1]https://github.com/AboubakerIdHamide/N7-SVM-vs-RF

II. Related Work

A. Machine Learning for Intrusion Detection

Machine learning-based intrusion detection systems have been extensively studied over the past decade. Buczak and Guven [1] provide a comprehensive survey of data mining techniques applied to intrusion detection, identifying Random Forest and SVM as the most effective approaches on classical benchmarks such as KDD Cup 99 and NSL-KDD. The authors note that ensemble methods generally outperform single classifiers due to their ability to reduce variance and capture complex decision boundaries.

Farnaaz and Jabbar [5] compared RF, SVM, and Extreme Learning Machine on the NSL-KDD dataset. Their results show that RF achieves 99.67% accuracy, superior to SVM's 98.89%. However, this evaluation considers neither inference latency nor behavior under feature reduction, limiting its practical applicability for real-time deployment scenarios.

B. Real-Time Constraints and Latency

The latency constraint for real-time IDS has been formalized by Ring et al. [4], who establish 50 ms as the upper acceptable limit for critical applications based on network response time requirements. This constraint has significant implications for model selection, as many high-accuracy models (deep neural networks, kernel SVMs with large support vector sets) exceed this threshold. The authors also provide a comprehensive review of IDS datasets, establishing CIC-IDS2017 as the current standard benchmark.

C. Robustness and Concept Drift

Concept drift represents a major challenge for production IDS, as attack signatures continuously evolve and new attack vectors emerge. Gama et al. [11] provide a foundational survey on concept drift adaptation, categorizing drift types and adaptation strategies. They note that models optimized for static distributions may fail catastrophically when the underlying data distribution changes.

Andresini et al. [7] proposed INSOMNIA, an incremental learning framework specifically designed to maintain IDS performance under drift through continuous model updates. Their results show that static models can lose up to 40% accuracy within months of deployment, highlighting the importance of robustness evaluation.

D. Positioning of Our Contribution

Unlike existing works, our study differs in three key aspects. First, we jointly measure precision (PR-AUC) and inference latency under the same 50 ms constraint, ensuring that reported models are deployable. Second, we quantify resilience to feature ablation via drastic reduction from 78 to 10 features, relevant for constrained deployment scenarios. Third, we simulate progressive concept drift to evaluate model robustness, revealing an undocumented tradeoff between precision and stability that has practical implications for model selection in production environments.

Table I
Class distribution in the CIC-IDS2017 subset (800K samples).

Class	Samples	%
BENIGN	646,038	80.8
DDoS	56,620	7.1
PortScan	55,462	6.9
DoS Hulk	33,369	4.2
DoS GoldenEye	4,521	0.6
FTP-Patator	2,890	0.4
SSH-Patator	1,100	0.1
Total Attacks	**153,962**	**19.2**

III. Methodology

A. Problem Formulation

Let $\mathcal{X} \subseteq \mathbb{R}^d$ denote the network feature space with $d = 78$ features and $\mathcal{Y} = \{0, 1\}$ the label space where 0 represents BENIGN traffic and 1 represents ATTACK traffic. Given a dataset $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$ with N samples, the objective is to learn a classifier $f : \mathcal{X} \to \mathcal{Y}$ that maximizes PR-AUC while satisfying the latency constraint:

$$\tau(f, \mathbf{x}) < 50 \text{ ms}, \quad \forall \mathbf{x} \in \mathcal{X} \tag{1}$$

where $\tau(f, \mathbf{x})$ denotes the inference time of classifier f on sample $\mathbf{x}$.

B. Dataset

Experiments use the CIC-IDS2017 dataset [2], a standard benchmark for intrusion detection created by the Canadian Institute for Cybersecurity. This dataset contains 2.8 million network flows captured over 5 days (Monday to Friday) in a controlled environment simulating realistic network conditions. Each flow is characterized by 78 features extracted via CICFlowMeter, including:

- **Flow-level features**: duration, packet counts, byte counts
- **Statistical features**: mean, std, min, max of packet sizes
- **Flag-based features**: TCP flag counts (SYN, FIN, RST, PSH, ACK)
- **Inter-arrival times**: forward and backward IAT statistics

The dataset includes 15 attack types across different days. Table I presents the class distribution for the subset used in this study (800,000 samples).

C. Preprocessing Pipeline

The preprocessing follows a strict anti-leakage protocol to ensure valid generalization estimates:

Step 1 — Data Cleaning:

- Removal of non-numeric columns (IP addresses, timestamps, flow IDs)
- Removal of constant-value features (zero variance)
- Replacement of infinite values with NaN

Step 2 — Train/Test Split:

- Stratified 80/20 split preserving class distribution
- Random seed fixed for reproducibility

Table II
TOP 10 FEATURES BY RANDOM FOREST GINI IMPORTANCE.

#	Feature	Imp.
1	Destination Port	.142
2	Init_Win_bytes_fwd	.098
3	Init_Win_bytes_bwd	.087
4	Bwd Pkt Len Max	.072
5	Fwd Pkt Len Max	.068
6	Flow Bytes/s	.054
7	Bwd Pkt Len Mean	.048
8	Avg Bwd Seg Size	.043
9	Fwd Pkt Len Mean	.039
10	Flow Duration	.035

Step 3 — Imputation and Scaling (fitted on training set only):

- Missing values replaced with median computed on training set
- StandardScaler normalization: $x' = (x - \mu_{train})/\sigma_{train}$

This protocol ensures that no information from the test set leaks into the training process, which would artificially inflate performance estimates.

D. Evaluated Models

Three experimental configurations are tested, each addressing a specific aspect of the research question:

Experiment 1 — Full feature comparison (78 features):

- **Random Forest** [8]: 100 trees, `class_weight=balanced`, `max_features=sqrt`, `n_jobs=-1` for parallel inference
- **LinearSVC** [9]: $C = 1.0$, `max_iter=2000`, `class_weight=balanced`, `dual=True`

Experiment 2 — Hybrid approach (14 features):

- Feature selection: top 14 features ranked by RF Gini importance
- Classification: SVM-RBF with $C = 1.0$, $\gamma = \text{scale}$

Experiment 3 — Ablation study (10 features):

- Reduction to first 10 features by RF importance ranking
- Comparison of RF vs SVM-RBF on reduced feature space

Table II presents the top 10 features identified by RF importance.

E. Concept Drift Simulation Protocol

To evaluate model robustness, concept drift is simulated via combined perturbation on the test set. This simulates the scenario where attack signatures evolve after model deployment [11]:

$$\mathbf{x}_i' = \mathbf{x}_i + \alpha \cdot \epsilon_i \cdot \sigma + \beta \cdot \delta \cdot \sigma \tag{2}$$

where $\epsilon_i \sim \mathcal{N}(0, 1)$ is Gaussian noise, δ is a distribution shift factor, σ is the feature-wise standard deviation, and α, β control the drift intensity.

Table III
EXPERIMENT 1: RF VS LINEARSVM (78 FEATURES). MEAN ± STD OVER $k = 5$ RUNS.

Model	PR-AUC	Recall	F1	Lat. (ms)
RF	.9997	.9972	.9968	36.3
LinearSVM	.9196	.9606	.8004	1.2

- Gaussian noise injection: $\alpha = 0.5 \times$ drift_level
- Distribution shift on 30% of features: $\beta = 0.5 \times$ drift_level
- Tested levels: 0%, 5%, 10%, 15%, 20%, 30%, 40%

F. Evaluation Metrics

Four complementary metrics are reported to provide a complete performance picture:

- **PR-AUC**: Area Under Precision-Recall Curve, computed using the average precision score. This metric is preferred over ROC-AUC for imbalanced datasets as it focuses on the minority class [3].
- **Recall**: $\frac{TP}{TP+FN}$, measuring the proportion of attacks correctly detected (sensitivity).
- **F1-Score**: $\frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision}+\text{Recall}}$, the harmonic mean providing a balanced measure.
- **Latency**: Mean inference time per sample in milliseconds, measured over 10 repetitions to account for system variability.

G. Statistical Validation Protocol

Each experiment is repeated $k = 5$ times with different random seeds (42, 123, 456, 789, 1011) affecting both train/test splits and model initialization. Results are reported as mean ± standard deviation following reproducibility best practices [12]. This protocol enables assessment of result stability and statistical significance without requiring computationally expensive k-fold cross-validation.

IV. EXPERIMENTAL RESULTS

This section presents factual results from the three experiments and robustness analysis. All interpretations are deferred to Section V.

A. Experiment 1: RF vs SVM (78 features)

Table III presents comparative results on all 78 features. Figure 1 visualizes the performance comparison.

RF achieves PR-AUC of 0.9997 ± 0.0000 versus 0.9196 ± 0.0016 for LinearSVM, a difference of 8.7 percentage points. Both RF latency (36.3 ± 10.3 ms) and LinearSVM latency (1.2 ± 0.4 ms) satisfy the 50 ms constraint.

B. Experiment 2: Hybrid Approach

The hybrid approach (RF feature selection + SVM-RBF classification on top 14 features) achieves PR-AUC of 0.8299 ± 0.0126, Recall of 0.9698 ± 0.0034, and F1-Score of 0.7439 ± 0.0089. Measured latency is 1.40 ± 0.42 ms. This configuration yields lower PR-AUC than both individual models.

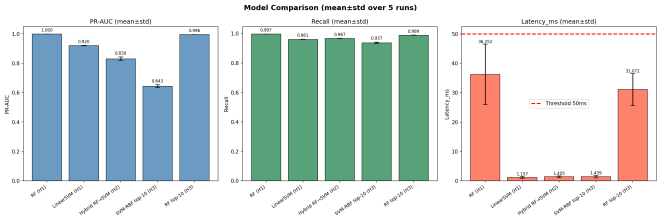

Figure 1. Performance comparison across models. Error bars represent standard deviation over $k = 5$ runs. Red line indicates 50 ms latency threshold.

Table IV
EXPERIMENT 3: ABLATION TO 10 FEATURES. MEAN $\pm$ STD OVER $k = 5$ RUNS.

Model	PR-AUC	Recall	F1	Lat. (ms)
RF top-10	.9962	.9890	.9823	31.1
SVM-RBF	.6434	.9372	.6917	1.4

Table V
F1-SCORE UNDER CONCEPT DRIFT (% CHANGE FROM BASELINE).

Drift	RF-78	LinSVM	RF-10
0%	.997	.799	.983
5%	.017 (–98%)	.736 (–8%)	.024 (–98%)
10%	.010 (–99%)	.623 (–22%)	.016 (–98%)
20%	.007 (–99%)	.498 (–38%)	.010 (–99%)
40%	.004 (–100%)	.412 (–48%)	.007 (–99%)

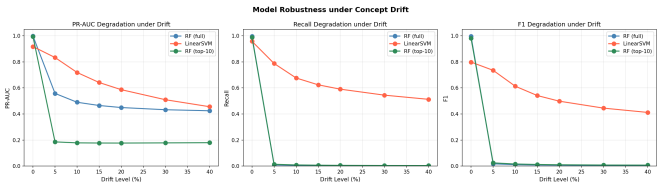

Figure 2. Metric evolution as function of concept drift level. RF exhibits sharp degradation at minimal drift while LinearSVM degrades progressively.

C. Experiment 3: Ablation Study (10 features)

Table IV presents results under ablation to 10 features.

RF top-10 achieves PR-AUC of 0.9962 ± 0.0002, representing 0.35% degradation from 78 features. SVM-RBF top-10 achieves PR-AUC of 0.6434 ± 0.0097, a 30% degradation from full features.

D. Robustness Analysis Under Drift

Table V presents F1-Score degradation under different concept drift levels. Figure 2 visualizes the degradation curves.

At 0% drift, RF achieves F1 of 0.9969 versus 0.7988 for LinearSVM. At 5% drift, RF drops to 0.0172 (F1 -98.3%) while LinearSVM reaches 0.7355 (F1 -7.9%). At 40% drift, RF achieves 0.0036 and LinearSVM 0.4120.

E. Precision-Recall Curves

Figure 3 presents the Precision-Recall curves for the evaluated models on static test data.

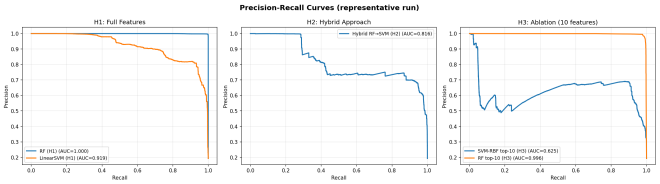

Figure 3. Precision-Recall curves on static test data. RF achieves near-perfect PR-AUC while SVM variants show varying precision at high recall levels.

Table VI
SUMMARY OF ALL EVALUATED CONFIGURATIONS.

Model	PR-AUC	F1	Lat.	<50ms
RF (78)	.9997	.9968	36 ms	Yes
LinearSVM	.9196	.8004	1 ms	Yes
Hybrid	.8299	.7439	1 ms	Yes
RF (10)	.9962	.9823	31 ms	Yes
SVM-RBF (10)	.6434	.6917	1 ms	Yes

F. Results Summary

Table VI synthesizes all experimental results.

V. DISCUSSION

A. Answer to Research Question

The results presented in Section IV address the formulated research question regarding model selection under latency constraints. RF maximizes PR-AUC (0.9997) under the 50 ms latency constraint on static data, outperforming LinearSVM by 8.7 percentage points. This superiority is attributable to RF's ability to capture non-linear feature interactions via multiple decision trees operating in parallel, unlike the linear SVM operating on a single hyperplane [8].

Regarding ablation resilience, RF maintains 99.6% of its original PR-AUC despite reduction from 78 to 10 features, while SVM-RBF drops by 30%. This property is attributable to the bagging mechanism that reduces variance and the implicit discriminative feature selection performed by individual trees [10]. The robustness to feature reduction makes RF suitable for deployment in resource-constrained environments.

The drift analysis reveals an unanticipated result: RF, optimal under static conditions, collapses under minimal perturbation (5% drift, F1 -98.3%). Conversely, LinearSVM, initially less accurate, maintains graceful degradation (F1 -7.9% at 5% drift). This behavior is explained by RF overfitting decision boundaries to the specific training distribution, creating brittle decision rules that fail when the data distribution shifts [11].

B. The Precision-Robustness Tradeoff

Results reveal an undocumented tradeoff: maximum precision (RF) accompanies fragility to distribution perturbations, while linear model simplicity (SVM) offers better stability at the cost of reduced precision. This tradeoff has not been previously documented in the IDS literature, where model comparisons typically assume static data distributions.

This tradeoff has direct practical implications for deployment:

- **Stable environments** with well-characterized traffic patterns: RF is optimal, achieving near-perfect detection with acceptable latency.
- **Evolving environments** with frequent concept drift due to new attack vectors: LinearSVM offers better operational stability, maintaining usable performance even under significant distribution shift.
- **Resource-constrained deployments** requiring dimensionality reduction: RF shows superior resilience to feature ablation.

C. Hybrid Approach Analysis

The hybrid approach (RF→SVM) achieves PR-AUC of 0.8299, inferior to both individual models. This counter-intuitive result is explained by two factors: (i) information loss during reduction to 14 features removes discriminative patterns that RF would have captured with full features, and (ii) SVM-RBF limitations on low-dimensional spaces where complex decision boundaries cannot be properly learned [9]. The hybrid approach fails to combine the strengths of both methods.

D. Practical Deployment Recommendations

Based on our findings, we provide the following recommendations for practitioners deploying ML-based IDS:

Environment assessment: Before model selection, organizations should assess the expected stability of their network traffic patterns. Networks with well-defined protocols and predictable traffic (e.g., industrial control systems) favor RF deployment, while dynamic environments (e.g., public-facing services) may benefit from LinearSVM's robustness.

Monitoring strategy: For RF deployments, continuous performance monitoring is essential. A sudden drop in detection rates may indicate concept drift, requiring model retraining or fallback to a more robust classifier.

Ensemble considerations: Organizations with sufficient computational resources might consider deploying both models in parallel, using RF for primary detection and LinearSVM as a fallback when drift is detected.

Feature engineering: Given RF's exceptional ablation resilience, organizations with constrained edge devices can confidently reduce features while maintaining detection capability. The top 10 features identified (Destination Port, Init_Win_bytes, packet lengths) provide a minimal viable feature set.

E. Limitations and Threats to Validity

This study presents several limitations requiring caution in generalizing conclusions:

Internal validity:

- Concept drift is artificially simulated via noise injection and distribution shift; validation on real evolving network data would strengthen conclusions.
- Absence of k-fold cross-validation limits statistical generalization; however, the 5-run protocol with different seeds provides variance estimation.
- The subsampling approach for SVM-RBF training (30,000 samples) may introduce bias compared to training on full data.

External validity:

- CIC-IDS2017, while a standard benchmark, was captured in 2017 and may not reflect current attack signatures and network traffic patterns characteristic of modern threats.
- Results are specific to the tested hyperparameter configurations; different tuning strategies (grid search, Bayesian optimization) may alter relative performance.
- Binary classification simplifies the multi-class nature of real IDS; per-attack-type analysis could reveal different model strengths.

Construct validity:

- The 50 ms latency threshold is based on literature recommendations [4] but may vary significantly by application domain and network infrastructure.
- Ablation study uses RF Gini importance ranking; alternative feature selection methods (mutual information, recursive elimination) could yield different results.

VI. Conclusion and Future Work

This study addressed the problem of real-time intrusion detection under strict latency constraints, comparing Random Forest and Support Vector Machine families across multiple evaluation dimensions: static detection performance, ablation resilience, and concept drift robustness.

A. Summary of Contributions

The research question aimed to identify the model family maximizing PR-AUC under the 50 ms constraint, and to evaluate resilience under feature ablation and concept drift. Our experimental evaluation on CIC-IDS2017 with statistical validation over $k = 5$ runs provides three main contributions to the IDS literature.

First, we provide a rigorous comparative evaluation with statistical validation, demonstrating that RF achieves superior static performance (PR-AUC 0.9997) while respecting the latency constraint, confirming its suitability for high-accuracy detection in stable environments.

Second, through our ablation study, we demonstrate that RF exhibits exceptional resilience to feature reduction, maintaining 99.6% of its performance with only 10 features. This finding has practical implications for resource-constrained deployments such as IoT gateways and edge devices where computational resources are limited.

Third, and most significantly, we identified and documented a previously unreported precision-robustness tradeoff: RF collapses under minimal concept drift (F1 -98.3% at 5% drift) while LinearSVM degrades gracefully (F1 -7.9%). This finding challenges the common assumption that the best-performing model on static benchmarks is optimal for production deployment.

B. Future Work

Several research directions emerge from our findings. First, **adaptive ensemble approaches** combining RF precision with online learning mechanisms warrant investigation. Techniques such as incremental learning or drift detection with model switching could maintain RF's accuracy while improving robustness.

Second, **validation on recent datasets** including CIC-IDS2018, UNSW-NB15, and real-world network captures with natural concept drift would strengthen the external validity of our conclusions.

Third, **ensemble diversity analysis** could explore whether deliberate diversification of RF base learners improves robustness to distribution shift without sacrificing static performance.

Finally, **multi-class evaluation** extending beyond binary classification to per-attack-type analysis could reveal attack categories where SVM outperforms RF, enabling attack-specific model selection strategies.

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